NeuroDrive Session Report

Participant: Alice #1 **Date:** 2026-05-23T20:15:26

Methodology

Preprocessing: notch filter at 50 Hz, bandpass 0.5–50 Hz, and ICA-based artifact removal per-window.

Spectral features: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (13–30 Hz), gamma (30–50 Hz).

Derived features: Frontal Alpha Asymmetry (FAA), Theta/Beta Ratio (TBR), Low-/High-Beta split (15–18 Hz / 19–22 Hz), Low/High Beta ratio, temporal asymmetry, and broadband PSD means.

Advanced features: differential entropy (DE), sample entropy (SampEn), and debiased phase-amplitude coupling (dPAC) between delta phase and high-beta amplitude.

Modeling: ensemble RandomForest trained with PCA dimensionality reduction and RFE-based feature selection; hyperparameters tuned via cross-validation.

Preprocessor: saved (2026-05-23T21:04:06) — PCA + RFE pipeline applied.
Model: RandomForestClassifier
Params: {'bootstrap': True, 'ccp_alpha': 0.0, 'class_weight': 'balanced', 'criterion': 'gini', 'max_depth': 12, 'max_features': 'sqrt'}
CV accuracy: 97.99% ± 0.19%
Top features: FAA (0.33), beta_frontal (0.33), beta_asym (0.14), theta (0.08), af7_a (0.05)

Ground-truth: anxiety labels are mapped to Low/Medium/High and aggregated per-session as mean score; for clinical use anchor to validated psychometric scales (STAI/HAM-A).

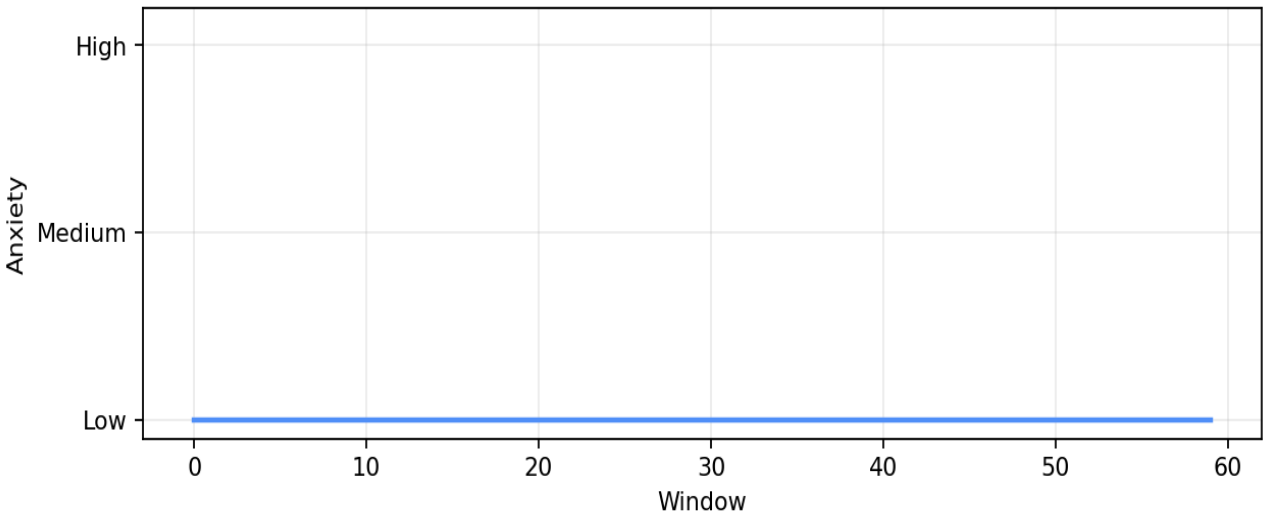
Methodology Excerpts

Frontal Alpha Asymmetry (FAA): FAA = ln(F4) - ln(F3). Alternatively: FAA = (F4 - F3)/(F4 + F3).

Debiased Phase-Amplitude Coupling (dPAC):\nPAC = (1/n) \u2211_t a_t e^{i\varphi_t}; mean phase clustering = (1/n) \u2211_t e^{i\varphi_t};\ndPAC = (1/n) \u2211_t |a_t e^{i(\varphi_t - \overline{PC})}|.

Duration	02:00
Average Anxiety	Low
Peak Anxiety	Low
Condition Tested	Baseline

Anxiety Timeline



Feature Averages

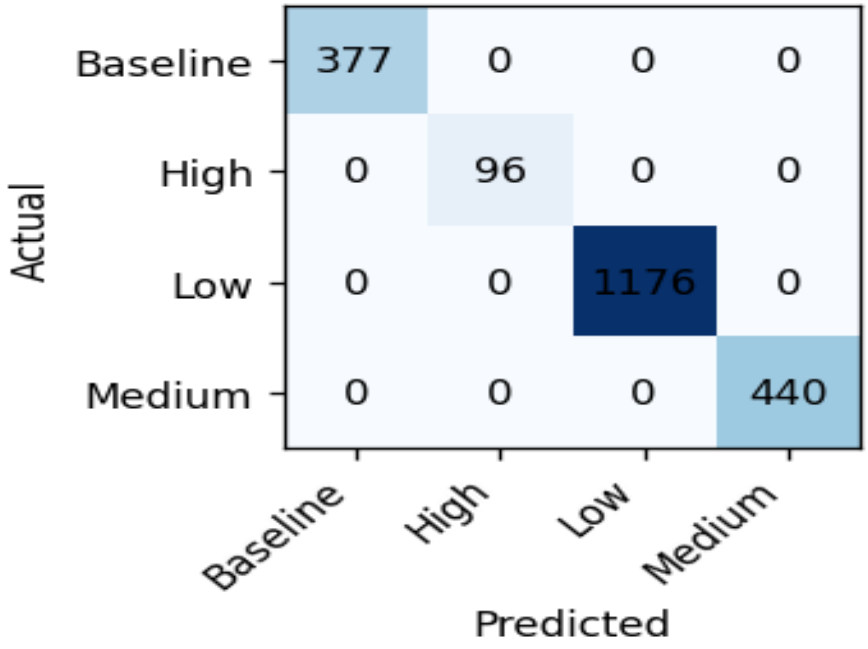
Feature	Average Value
FAA	-0.336
beta_frontal	1.204
low_beta_frontal	0.000
high_beta_frontal	0.000
low_high_beta_ratio	0.000
beta_asym	-0.133
theta	0.813
TBR	0.666
alpha_front	0.593
temp_asym	0.106
af7_a	0.509
af8_a	0.575
af7_b	1.032
af8_b	1.104
delta	0.331
theta_band	0.772
alpha_band	0.620
beta_band	1.190
gamma_band	0.188
DE_frontal	0.000
SampEn_frontal	0.000
dPAC_frontal	0.000
HR	0.000

Top 5 Peak Anxiety Moments

Timestamp	Anxiety
00:00	Low
00:02	Low
00:04	Low
00:06	Low
00:08	Low

Model Performance

Class	Precision	Recall	F1-score	Support
Baseline	1.00	1.00	1.00	377
High	1.00	1.00	1.00	96
Low	1.00	1.00	1.00	1176
Medium	1.00	1.00	1.00	440



Intervention Suggestions

Average anxiety was low. Provide positive reinforcement and continue regular monitoring.

References

Al-Ezzi, A., Kamel, N., Faye, I., & Gunaseli, E. (2020). Review of EEG, ERP, and brain connectivity estimators as predictive biomarkers of social anxiety disorder. Frontiers in Psychology. Retrieved May 21, 2026, from https://www.frontiersin.org/articles/10.3389/fpsyg.2020.00730/full

Geetha, A., Jackulin Mahariba, A., Selvi, S., Kumaresan, T., Zulaikha Beevi, S., & Saranya, K. (2025). EEG-based anxiety and stress detection using PCA-RFE hybrid feature extraction and XGForest classification with hyperparameter optimization. TPM. Retrieved May 21, 2026, from https://tpmap.org/submission/index.php/tpm/article/view/473

Kesebir, S., Demirer, R. M., & Tarhan, N. (2019). Cross-frequency coupling (delta–beta) related with mixed features and response to treatment in bipolar II depression. PMC. Retrieved May 21, 2026, from

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